Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

EIN 4360 Facilities Planning and Material Handling

Module 12– Facility Location Models: Minimax

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POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

Single-Facility Location Model

- **Minisum Location Problem**
  - Minimizing the sum of the weighted distance between the new facility and other existing facilities
- **Minimax Location Problem**
  - Minimizing the maximum distance between the new facility and any existing facility

Single-Facility Location Model Examples

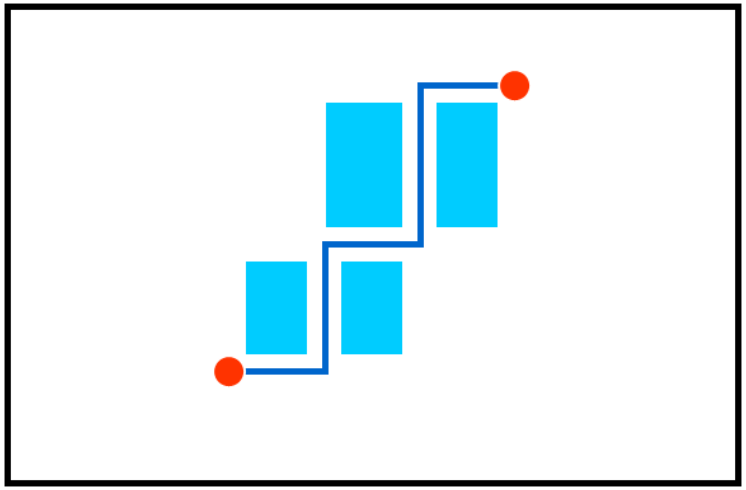
- **Minisum Location Problem**
  - Consider Minisum, when the weights of different locations are important.
  - For example:
    - Travel frequency
    - Cost
    - Flow
    - Probability of demand
    - Or some combination of these weights

Single-Facility Location Model Examples

- **Minimax Location Problem**
  - Consider Minimax, when only the furthest facility must be considered
  - For example:
    - Emergency facilities
    - Restrooms
    - Cafeterias

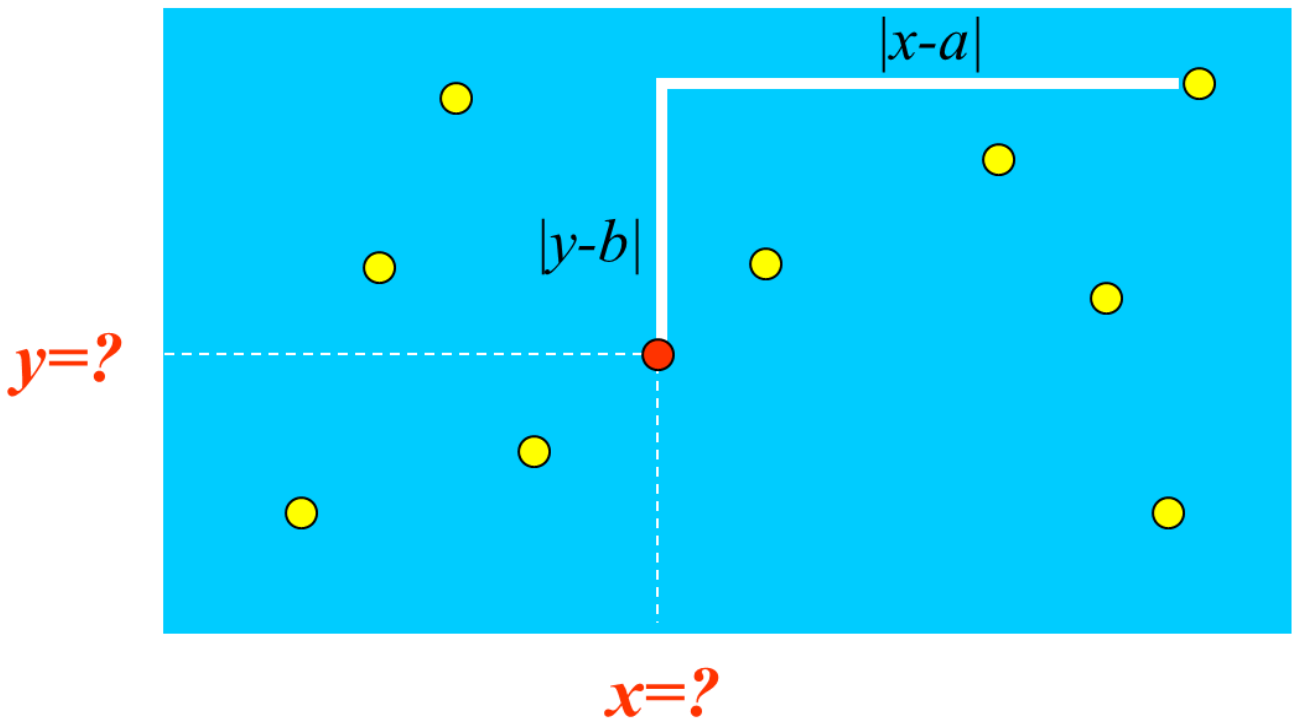
Distances

- **Rectilinear distances**



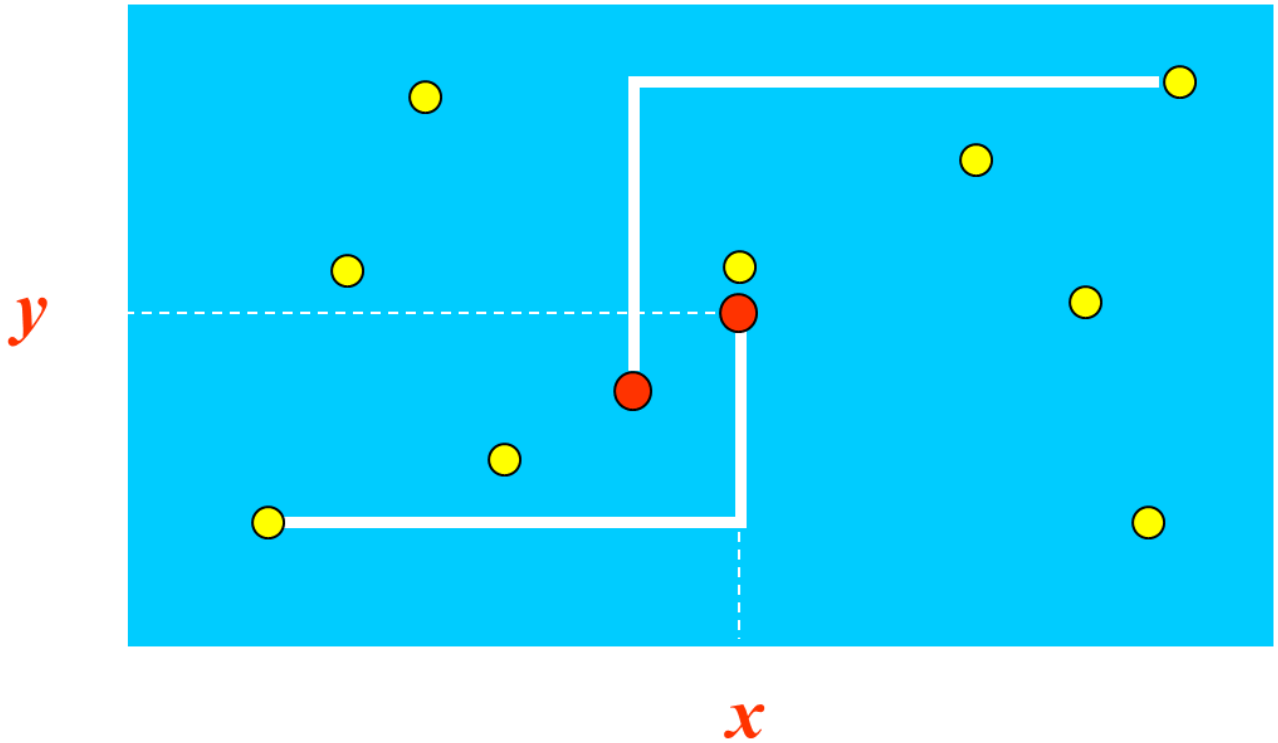
Distances

- Existing facility
- New facility



Distances

Place new facility so that longest rectangular distance will be as low as possible



Minimax Location Problem with Rectilinear Distances

Minimize maximum distance

$$Min \left\{ z(x, y) = \underset{i = 1,\ldots,m}{Max} \left(|x - a_i| + |y - b_i| \right) \right\}$$

$$Min \left\{ z(x, y) = Max \left(|x - a_1| + |y - b_1|, |x - a_2| + |y - b_2|, \ldots, |x - a_m| + |y - b_m| \right) \right\}$$

Minimax Location Problem with Rectilinear Distances

Formulas

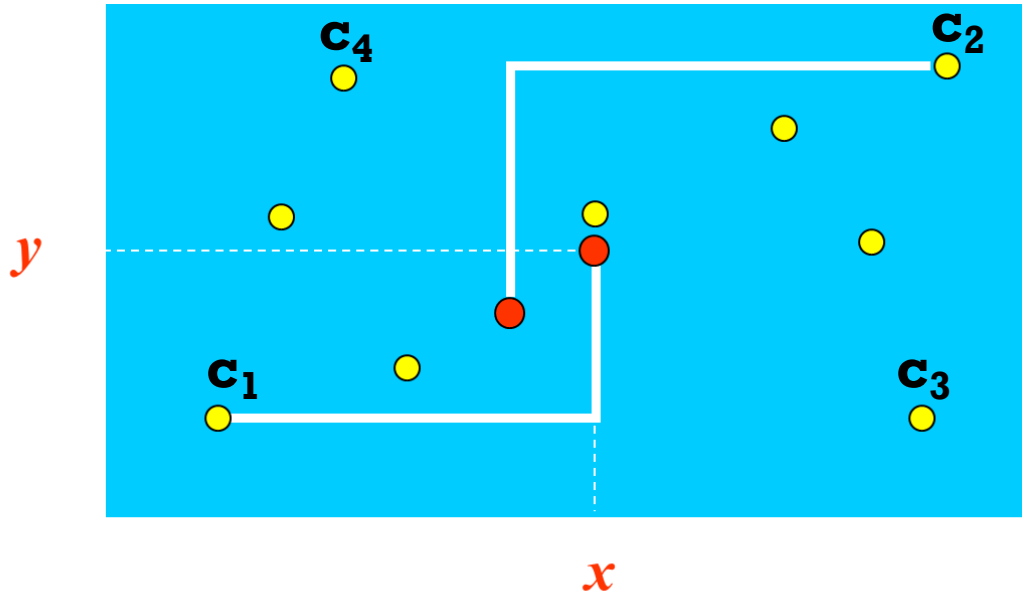
$c_1 = min\ (a_i + b_i)$ **Bottom-left**

$c_2 = max\ (a_i + b_i)$ **Top-right**

$c_3 = min\ (-a_i + b_i)$ **Bottom-right**

$c_4 = max\ (-a_i + b_i)$ **Top-left**

$c_5 = max\ (c_2 - c_1,\ c_4 - c_3)$



Minimax Location Problem with Rectilinear Distances

$c_1 = min\ (a_i + b_i)$ **Bottom-left**

$c_2 = max\ (a_i + b_i)$ **Top-right**

$c_3 = min\ (-a_i + b_i)$ **Bottom-right**

$c_4 = max\ (-a_i + b_i)$ **Top-left**

$c_5 = max\ (c_2 - c_1,\ c_4 - c_3)$

Formulas

Optimal Solution:

Line segment joining the points:

$$A(x^*, y^*) = \tfrac{1}{2}(c_1 - c_3,\ c_1 + c_3 + c_5)$$

and

$$B(x^*, y^*) = \tfrac{1}{2}(c_2 - c_4,\ c_2 + c_4 - c_5)$$

Maximum distance:

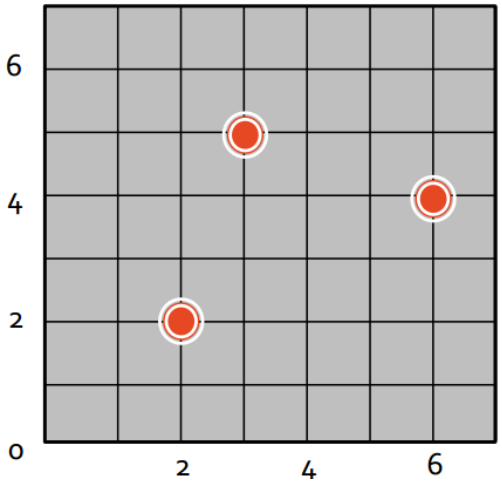
$$z^* = c_5 / 2$$

Example

Suppose we need to locate a fire station to serve **three** subdivisions of a city located at coordinates (2, 2) (6, 4), and (3,5) in miles. What is the optimal location for the fire station to **minimize the maximum distance** from the fire station to each subdivision?

a) What is the **optimal location** for the fire station to minimize the maximum distance from the fire station to each subdivision?

b) What is the **maximum distance** from the fire station to the furthest away subdivision?



Example

$c_1 = min\ (a_i + b_i)$ **Bottom-left**

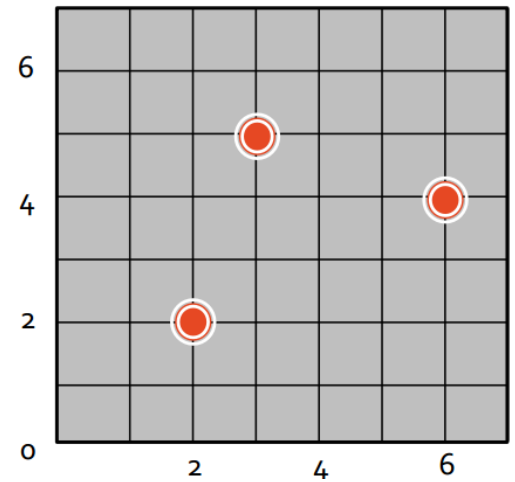
$c_2 = max\ (a_i + b_i)$ **Top-right**

$c_3 = min\ (-a_i + b_i)$ **Bottom-right**

$c_4 = max\ (-a_i + b_i)$ **Top-left**

$c_5 = max\ (c_2 - c_1,\ c_4 - c_3)$

Max distance between either diagonal



Optimal Solution:

Line segment joining the points:

$A(x^*, y^*) = \frac{1}{2}(c_1 - c_3,\ c_1 + c_3 + c_5)$

and

$B(x^*, y^*) = \frac{1}{2}(c_2 - c_4,\ c_2 + c_4 - c_5)$

Maximum distance:

$z^* = c_5 / 2$

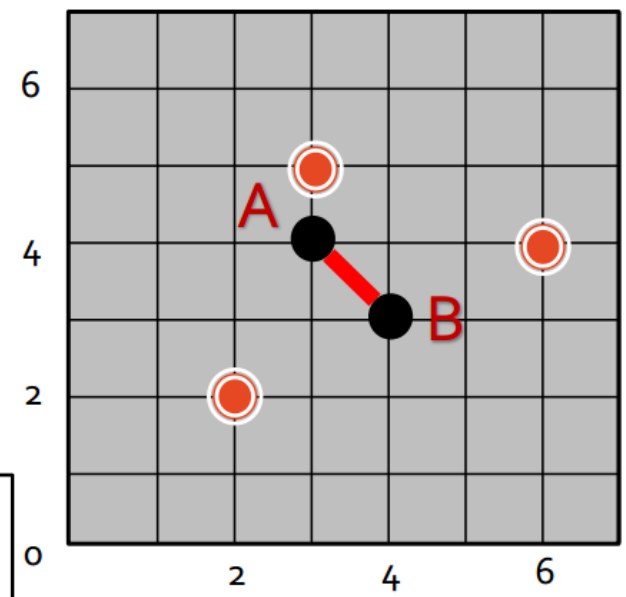
Example: Solution

$c_1 = min\ (a_i+b_i)$
$c_2 = max\ (a_i+b_i)$
$c_3 = min\ (-a_i+b_i)$
$c_4 = max\ (-a_i+b_i)$
$c_5 = max\ (c_2 - c_1,\ c_4 - c_3)$

$c_1 = min\ (\mathbf{2+2}, 3+5, 6+4)=4$
$c_2 = max\ (2+2, 3+5, \mathbf{6+4})=10$
$c_3 = min\ (-2+2, -3+5, \mathbf{-6+4})=-2$
$c_4 = max\ (-2+2, \mathbf{-3+5}, -6+4)=2$
$c_5 = max\ (10-4,\ 2-(-2))=6$

Optimal solution: *Lies on the line segment between A and B. In this case at* ***point A*** *is an attractive solution because this is also the location of the minisum.*

Optimal solution:
Line segment joining the points:
$A(x^*, y^*) = \frac{1}{2}(c_1 - c_3,\ c_1+c_3+c_5)$
$B(x^*, y^*) = \frac{1}{2}(c_2 - c_4,\ c_2 + c_4 - c_5)$
Maximum distance: $z^* = c_5/2$



Optimal solution:
Line segment joining the points:
$A(x^*, y^*) = \frac{1}{2}(4-(-2),\ 4+(-2)+6)=(3,4)$
$B(x^*, y^*) = \frac{1}{2}(10-2,\ 10+2-6)=(4,3)$
Maximum distance: $z^* = 6/2=3$

Example: LP Formulation

Locate an emergency service unit to serve the following sub-divisions of a city:

Subdivision	a_i	b_i
A	4	3
B	5	11
C	13	13
D	10	6
E	4	6
F	10	10
G	14	2

Example: LP Formulation

Minimize z

Subject to:

$$x+y-z \leq c_1$$

$$x+y+z \geq c_2$$

$$-x+y-z \leq c_3$$

$$-x+y+z \geq c_4$$

$$x, y, z \geq 0$$

Example: LP Formulation

Minimize z

Subject to:

$$x+y-z \leq 7$$

$$x+y+z \geq 26$$

From our example:

$$-x+y-z \leq -12$$

$$-x+y+z \geq 6$$

$$x, y, z \geq 0$$

Example: LP Formulation

```
Minimize  z
s.t.
   x + y - z <= 7
   x + y + z >= 26
  -x + y - z <= -12
  -x + y + z >= 6
END
```

```
OBJECTIVE FUNCTION VALUE
     1)      9.500000
VARIABLE                VALUE          REDUCED COST
     Z                  9.500000          0.000000
     X                  10.000000         0.000000
     Y                  6.500000          0.000000
   ROW             SLACK/SURPLUS     DUAL PRICES
     2)                 0.000000          0.500000
     3)                 0.000000         -0.500000
     4)                 1.000000          0.000000
     5)                 0.000000          0.000000
NO. ITERATIONS=         3
```

Contour corresponding to z

$$x+y \leq 7+z$$

$$x+y \geq 26-z$$

$$-x+y \leq -12+z$$

$$-x+y \geq 6-z$$

Contour corresponding to z=12

$$x+y \leq 7+12 \qquad x+y \leq 19$$

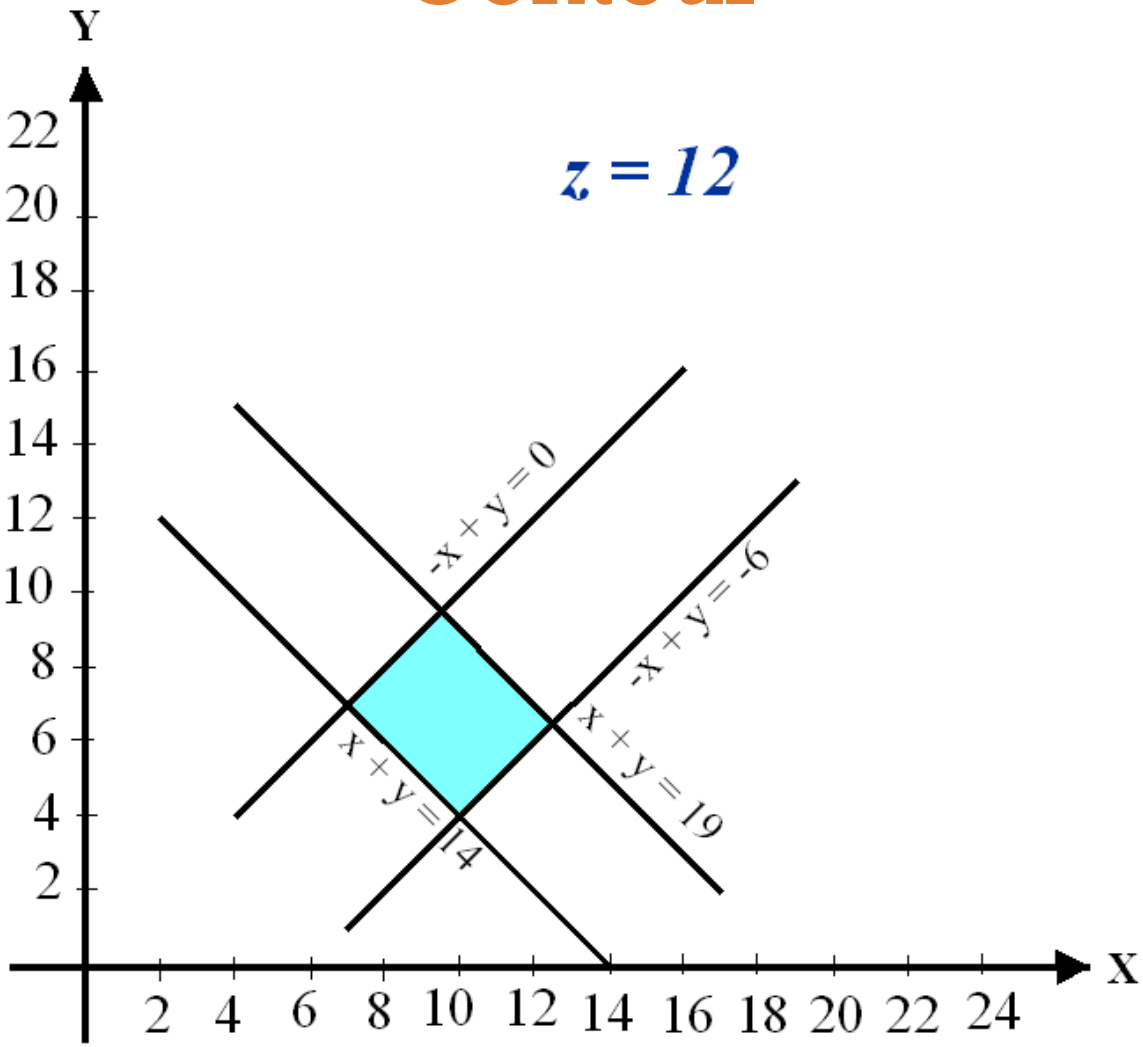
$$x+y \geq 26-12 \qquad x+y \geq 14$$

$$\longrightarrow$$

$$-x+y \leq -12+12 \qquad -x+y \leq 0$$

$$-x+y \geq 6-12 \qquad -x+y \geq -6$$

Contour



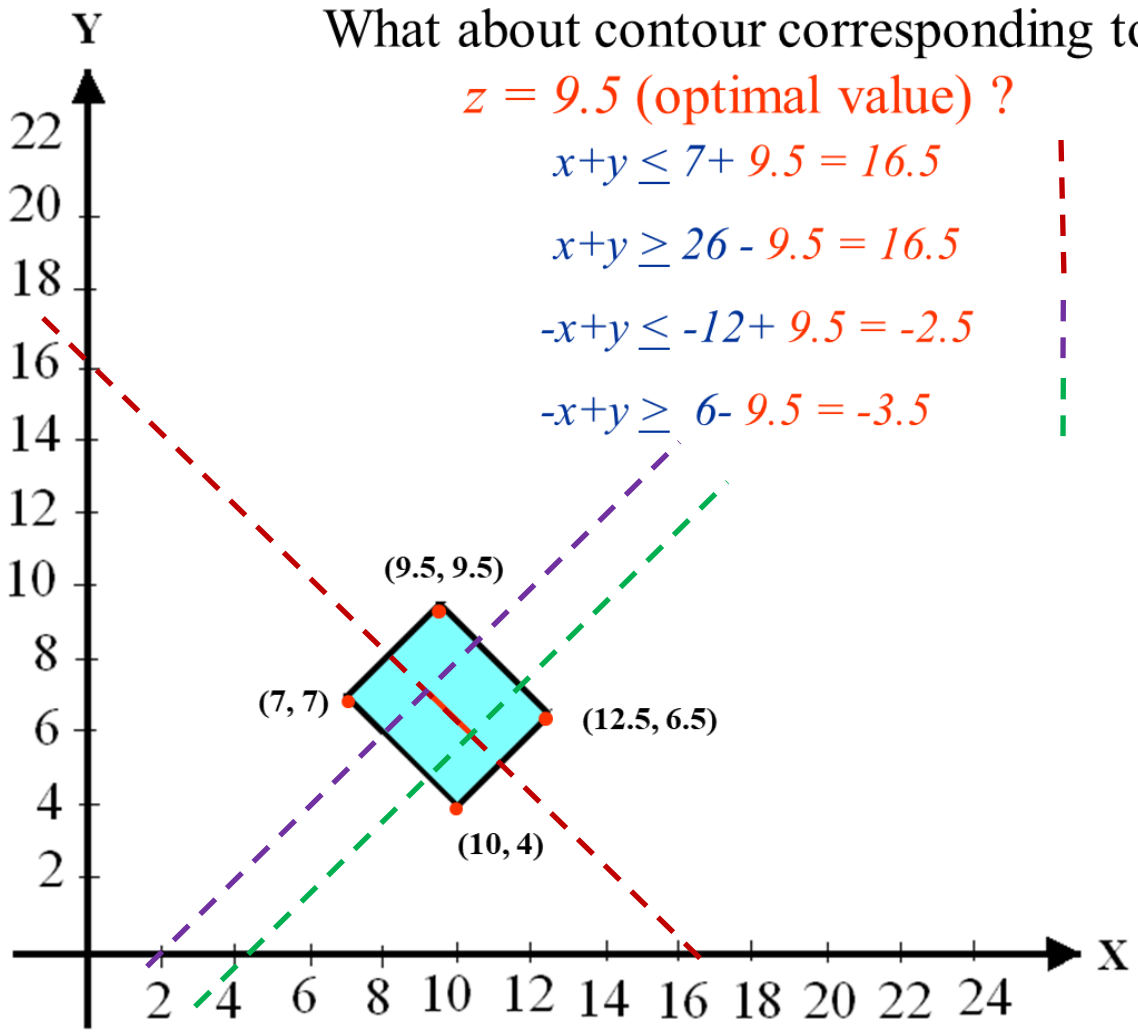
Contour

What about contour corresponding to

$z = 9.5$ (optimal value) ?

$x+y \leq 7+ 9.5 = 16.5$

$x+y \geq 26 - 9.5 = 16.5$

$-x+y \leq -12+ 9.5 = -2.5$

$-x+y \geq 6- 9.5 = -3.5$